Circle the correct answer for each question. If you're unsure, you can write numbers 1, 2, 3, etc. next to the choices to get partial credit.

Suppose a random sample of 100 students is taken at a university, asking the students how many hours they slept last night. The sample has a mean of 7.1 hours and a standard deviation of 1.2 hours. A 95% confidence interval is then constructed for the mean number of hours all students from that university slept last night. Use this to answer the following three questions:

1. Suppose, instead, that a 90% confidence interval is constructed with the data. How would the 90% confidence interval compare to your 95% confidence interval? The 90% confidence interval would be:

   (a) Wider
   (b) Same width
   (c) Cannot be determined with the information provided
   (d) Narrower

2. Suppose another random sample is taken with 150 students, and has the same sample mean and standard deviation as the original sample. How would the new 95% confidence interval compare to the original? The the new interval would be:

   (a) Wider
   (b) Narrower
   (c) Same width
   (d) Cannot be determined with the information provided

3. Suppose another sample of 100 students is taken and this time the mean is 7.1 with a standard deviation of 1.4 for the hours slept. These new statistics are used to create another 95% confidence interval for the mean hours slept. How would the new 95% confidence interval compare to the original? The new 95% confidence interval would be:

   (a) Wider
   (b) Narrower
   (c) Same width
   (d) Cannot be determined with the information provided

4. Researchers are studying a new weight-loss program. Using a sample of 9,812 volunteers with a desire to lose weight, a 95% confidence interval is constructed for the true mean amount of weight loss after six months on the program, and found to be (0.12, 0.20) pounds.
   Based on this interval, does statistically significant evidence exist that the weight-loss program works?

   (a) The question cannot be answered without the p-value.
   (b) Yes, since the entire interval is positive.
   (c) No, since the true mean amount of weight loss after six months on the program is less than a pound.
   (d) Yes, since the sample size was large.

5. A 2012 survey of 1,254 American cell phone owners indicates that 17% of those surveyed do browsing on their phone rather than a computer or other device.
   Suppose another survey was done on 1,254 American cell phone owners, but this time, 23% of cell phone owners surveyed do browsing on their phone rather than a computer or other device. How would the 95% confidence interval from this survey compare to the original? The new survey's interval would be:

   (a) Same width
   (b) Cannot be determined with the information provided
   (c) Wider
   (d) Narrower

6. Suppose data is collected on the heights of all students at a college. Each of the 250 people taking a statistics course randomly takes a sample of 40 of these heights and constructs a 95% confidence interval for mean height of all students at the college.
   Which of the following statements about the confidence intervals is most accurate?

   (a) About 95% of the time, a student's sample mean height will be contained in the interval the student found.
   (b) About 95% of the intervals will contain the population mean height.
   (c) About 95% of the heights of all students at the college will be contained in these intervals.
   (d) About 95% of the intervals will be identical.

Researchers are interested in investigating the effect of a drug used in the treatment of patients with glaucoma (an eye disorder associated with high eye pressure). In a volunteer sample of 35 patients with glaucoma in both eyes, one eye of each patient was randomly assigned to this drug, and the other eye was given a placebo. After one week, the eye pressure was measured on each eye. The difference in eye pressure between the two eyes (drug – placebo) was measured for each patient. The sample mean difference in eye pressure was –1.21 mmHg (millimeters of mercury), and the sample standard deviation of the differences was 4.67 mmHg.
Use this situation to answer the next two questions.

7. Use the 2SD method to find a 95% confidence interval for the population mean difference in eye pressure (drug – placebo).

   (a) (-2.00, -0.42)
   (b) Not enough information exists.
   (c) (-2.51, 0.09)
   (d) (-2.79, 0.37)

8. Suppose a 95% confidence interval for the population mean difference in eye pressure is (L, U). How would you interpret this interval?

   (a) In the long run, 95% of samples will produce a confidence interval of L to U mmH.
   (b) Approximately 95% of patients will have a difference in eye pressure (drug – placebo) between L and U mmHg.
   (c) 95% confidence exists that the population mean difference in eye pressure (drug – placebo) if treated would be between L and U mmHg.
   (d) 95% confidence exists that each patient's difference in eye pressure (drug – placebo) will be between L and U mmHg.

9. A 95% confidence interval is computed to estimate the mean household income for a city. Which of the following values will definitely be within the limits of this confidence interval?

   (a) The sample standard deviation
   (b) The sample mean
   (c) The p-value
   (d) The population mean